

Serie 17

DETERMINANT

1. Compute the determinant of

$$B = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 2 & 3 \\ 0 & 0 & 2 & 3 & 0 \\ 0 & 2 & 3 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 \end{pmatrix}$$

over $\mathbb{R}$ and over $\mathbb{F}_5$. Is B invertible?

2. (a) Let K be a field, let $\lambda \in K$, and let $A \in M_{n \times n}(K)$. Show that:
- i. For B such that $A \xrightarrow{\lambda L_i \rightarrow L_i} B$, $\det B = \lambda \det A$;
 - ii. For B such that $A \xrightarrow{L_i \leftrightarrow L_j} B$, $\det B = -\det A$;
 - iii. For B such that $A \xrightarrow{\lambda L_i + L_j \rightarrow L_j} B$ with $i \neq j$, $\det B = \det A$.
- (b) The integers 2014, 1484, 3710 and 6996 are all multiples of 106. Show (without brute-force calculations) that also

$$\det \begin{pmatrix} 2 & 1 & 3 & 6 \\ 0 & 4 & 7 & 9 \\ 1 & 8 & 1 & 9 \\ 4 & 4 & 0 & 6 \end{pmatrix}$$

is a multiple of 106.

3. Compute the determinants of the following matrices

$$A = \begin{pmatrix} 0 & 0 & 0 & 1 \\ -1 & 2 & 0 & 1 \\ 1 & 2 & -3 & 1 \\ 0 & -4 & 2 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix},$$

$$C = \begin{pmatrix} 2 & -3 & 5 & 1 & 4 \\ 2 & -3 & 1 & -6 & 18 \\ 4 & -3 & 9 & 6 & 10 \\ -2 & 4 & -6 & -1 & -1 \\ -6 & 11 & -23 & -14 & 9 \end{pmatrix}.$$

4. Let $A_n \in M(n \times n, \mathbb{R})$ be the matrix

$$\begin{pmatrix} 2 & 1 & 0 & \dots & 0 \\ 1 & 2 & 1 & \dots & 0 \\ 0 & \ddots & \ddots & \ddots & \vdots \\ \vdots & & & 1 & 2 & 1 \\ 0 & \dots & 0 & 1 & 2 \end{pmatrix}$$

Prove that

$$\det(A_n) = n + 1.$$

5. Let K be a field. If A is a 2×2 matrix over K , the classical adjoint of A is the 2×2 matrix $\text{adj } A$ defined by

$$\text{adj } A = \begin{pmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{pmatrix}$$

If $\det$ denotes the unique determinant function on 2×2 matrices over K , show that

- (a) $(\text{adj } A)A = A(\text{adj } A) = (\det A)I$;
- (b) $\det(\text{adj } A) = \det(A)$;
- (c) $\text{adj}(A^t) = (\text{adj } A)^t$.

(A^t denotes the transpose of A .)

6. (a) List explicitly the 24 permutations of degree 4, state which are odd and which are even, and use this to give the complete Leibniz formula

$$\det(A) = \sum_{\sigma} (\text{sgn } \sigma) A(1, \sigma 1) \cdots A(n, \sigma n)$$

for the determinant of a 4×4 matrix. Notice that for $n \geq 4$ it is not sufficient to compute a combination of the diagonals of a matrix to obtain its determinant.

- (b) For a general $n \in \mathbb{N}_{\geq 1}$, how many *even* permutations are there in S_n ?

7. An $n \times n$ matrix A is called triangular if $A_{ij} = 0$ whenever $i > j$ or if $A_{ij} = 0$ whenever $i < j$. Prove that the determinant of a triangular matrix is the product $A_{11}A_{22} \cdots A_{nn}$ of its diagonal entries.

8. Let $n \in \mathbb{N}_{\geq 2}$. Show that

$$\det \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix} = \prod_{1 \leq i < j \leq n} (x_j - x_i).$$

Remark. Products of the sort are called a *Vandermonde determinants* and the above matrix is called a *Vandermonde matrix*.

9. Let K be a field and let $A, B, C, D \in M_{n \times n}(K)$. Assume that A and C commute and that $\det A \neq 0$. Show that

$$\det \left(\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right) = \det(A \cdot D - C \cdot B)$$

Hint. Consider the matrix

$$\left(\begin{array}{c|c} I_n & O_n \\ \hline -C & A \end{array} \right).$$

10. Prove the following proposition using the Leibniz formula for determinants:

Proposition. *Let K be a field, and let $A, B \in M_{n \times n}(K)$. Then*

$$\det(AB) = \det(A) \cdot \det(B).$$

Hint. Denote $\{\mathbf{e}_i\}_{i=1}^n$ the standard basis of K^n and write the matrix B as a list of column blocks:

$$B = \left(\sum_{s_1=1}^n B(s_1, 1)\mathbf{e}_{s_1} \mid \cdots \mid \sum_{s_n=1}^n B(s_n, n)\mathbf{e}_{s_n} \right).$$

You will also need to prove the following lemma

Lemma. *For any $A \in M_{n \times n}(K)$ and any $\sigma \in S_n$, we have*

$$\det \left(A \cdot \mathbf{e}_{\sigma(1)} \mid \cdots \mid A \cdot \mathbf{e}_{\sigma(n)} \right) = \text{sgn}(\sigma) \det(A).$$

Single Choice. In each exercise, exactly one answer is correct.

1. For which $x \in \mathbb{R}$ do we have $\det \begin{pmatrix} 1 & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{pmatrix} = 1$?
 - (a) $x = -2$
 - (b) $x = 2$
 - (c) $x = -1$
 - (d) $x = 1$
2. Let K be a field and $\text{Mat}_{n \times n}(K)$ the vector space of $n \times n$ -Matrizen over K . Which assertion is wrong in general?
 - (a) A matrix $A \in \text{Mat}_{n \times n}(K)$ over K is invertible if and only if $\det(A) \neq 0$.
 - (b) The determinant of an upper triangular matrix only depends on its diagonal entries.
 - (c) For every $n \geq 0$ the determinant is a linear map $\text{Mat}_{n \times n}(K) \rightarrow K$.
 - (d) For every $n > 0$ the determinant map $\text{Mat}_{n \times n}(K) \rightarrow K$ is surjective.
3. In general, which operation changes the determinant?
 - (a) Exchanging two rows.
 - (b) Adding the multiple of one row to another
 - (c) Transpose.
 - (d) Replacement by similar matrix.

Multiple Choice Fragen.

1. Which of the following assertions are correct for arbitrary $A, B \in M_{n \times n}(\mathbb{R})$ with $n \geq 2$?
 - (a) We have $\det(AB) = \det(A) \det(B)$.
 - (b) From $\det(A) \neq 0$ it follows that the column vectors $a_1, \dots, a_n$ of A are linearly independent.
 - (c) Es gilt $\det(AB) = \det(BA)$.
 - (d) For every non-zero real number λ we have $\det(\lambda A) = \lambda \det(A)$.
 - (e) We have $\det(A + B) = \det(A) + \det(B)$.
2. Let $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = 4$. Which of the following statements are true?
 - (a) $\det \begin{pmatrix} 2a & 2b \\ 2c & 2d \end{pmatrix} = 8$.
 - (b) $\det \begin{pmatrix} a & b \\ c - a & d - b \end{pmatrix} = 4$.
 - (c) $\det \begin{pmatrix} a & b \\ c + 2a & d + 2b \end{pmatrix} = 4$.
 - (d) $\det \begin{pmatrix} a & b \\ 3c & 3d \end{pmatrix} = 12$.